

1.5 Foundation Checklist

Pre-pour preparation, pour-day execution, and post-pour finishing — every step from batter boards to backfill.

Project Name: _____ Foundation Type: _____

Owner-Builder: _____ Start Date: _____

Foundation Contractor: _____ Target Completion: _____

INSTRUCTIONS: This checklist covers all phases of foundation construction: pre-pour preparation, pour day activities, and post-pour finishing. A proper foundation is critical — take your time, double-check measurements, and don't skip inspections. Document everything with photos.

⚠ CRITICAL REMINDER — Foundation work cannot proceed until:

- Building permit is approved and posted on site
- Site preparation is 100% complete
- All utilities are located (811 call completed)
- Foundation layout has been verified and approved

SECTION A: PRE-POUR PREPARATION

A1: Foundation Layout & Batter Boards

	Task	Date	Notes / Measurements
☐	Review foundation plans and verify all dimensions		
☐	Establish reference point/benchmark (permanent marker, stake, etc.)		
☐	Set up batter boards 3-4 feet outside all corners		
☐	String layout lines between batter boards		
☐	Verify all corners are square (3-4-5 method or diagonal measurement)		
☐	Diagonal measurements (must be equal!): A: _____ B: _____		

A1: Foundation Layout & Batter Boards

	Task	Date	Notes / Measurements
☐	Mark excavation limits with spray paint or stakes		
☐	Take photos of string layout from multiple angles		
☐	Have layout verified by surveyor or experienced builder (recommended)		

A2: Excavation

	Task	Date	Notes / Measurements
☐	Verify frost depth requirement (check local code): Frost depth (in): _____		
☐	Excavate to required depth below finish grade (frost depth + footer)		
☐	Verify excavation depth at all corners and mid-points (record below)		
☐	NE: _____ NW: _____ SE: _____ SW: _____		
☐	Remove all loose soil, roots, organic matter from excavation		
☐	Ensure excavation bottom is level and undisturbed soil		
☐	Document any rock, ledge, or unusual soil conditions encountered		
☐	Stockpile excavated soil away from work area (for later backfill)		

A3: Gravel Base & Compaction

	Task	Date	Notes / Measurements
☐	Install gravel base per plan (typically 4-6" of crushed stone)		
☐	Gravel type used: _____ Depth (in): _____		
☐	Compact gravel base with plate compactor or roller		

A3: Gravel Base & Compaction

	Task	Date	Notes / Measurements
☐	Verify base is level (max 1/4" variance in 10 feet)		
☐	Install drainage tile around perimeter (if required)		
☐	Verify drainage tile slopes to daylight or sump (1/8" per foot min)		

A4: Footer Forms & Rebar

	Task	Date	Notes / Measurements
☐	Install footer forms per plan dimensions		
☐	Footer width (in): _____ Footer depth (in): _____		
☐	Forms are straight, level, and properly braced		
☐	Verify footer width and depth at multiple locations		
☐	Install rebar per plan (typically #4 or #5, 2 bars continuous)		
☐	Rebar size: # _____ Configuration: _____		
☐	Rebar is elevated on chairs (3" min clearance from bottom)		
☐	All rebar splices overlap minimum 18" and are tied with wire		
☐	Rebar is clean (no rust scale, oil, or debris)		
☐	Corner rebar is properly bent or overlapped (no gaps)		

A5: Plumbing Under-Slab Rough-In

	Task	Date	Notes / Measurements
☐	Install all plumbing stubs per plan (water supply, drains, vents)		
☐	Verify plumbing stub locations match plan dimensions		

A5: Plumbing Under-Slab Rough-In

	Task	Date	Notes / Measurements
☐	All drain pipes slope properly (1/4" per foot minimum)		
☐	Pressure test water supply lines (if required by inspector)		
☐	Cap all plumbing stubs to prevent concrete entry during pour		
☐	Support all pipes properly (no sagging, proper bedding in gravel)		
☐	Take photos of all plumbing locations before pour		

A6: Vapor Barrier (for slab-on-grade)

	Task	Date	Notes / Measurements
☐	Install vapor barrier (minimum 6 mil poly, 10 mil preferred)		
☐	Overlap all seams minimum 6" and tape with approved sealing tape		
☐	Run vapor barrier up sides of forms (3-6" above slab height)		
☐	Seal vapor barrier around all plumbing penetrations		
☐	Repair any tears or punctures with approved patch/tape		
☐	Place sand layer over vapor barrier if required (prevents punctures)		

A7: Foundation Wall Forms (if applicable)

	Task	Date	Notes / Measurements
☐	Erect foundation wall forms (ICF, wood, or metal forms)		
☐	Wall height (in): _____ Wall thickness (in): _____		
☐	Forms are plumb, level, and properly braced (check with 4' level)		

A7: Foundation Wall Forms (if applicable)

	Task	Date	Notes / Measurements
☐	Install tie rods/snap ties at proper spacing per form manufacturer		
☐	Install vertical rebar in walls per plan (typically #4 @ 16-24" o.c.)		
☐	Install horizontal rebar per plan (min 2 continuous bars)		
☐	Tie vertical rebar to footer rebar (dowels or epoxy anchors)		
☐	Install anchor bolts at proper spacing (typically 6' o.c., 12" from corners)		
☐	Anchor bolt size: _____ Embedment depth (in): _____		
☐	Create forms for window wells, bulkhead openings (if applicable)		
☐	Coat forms with release agent/oil (prevents concrete adhesion)		
☐	Double-check all dimensions before calling for inspection		

A8: Pre-Inspection Checklist

Critical items to verify before calling the inspector:

- ☐ Foundation layout is square (diagonals match within 1/4")
- ☐ Excavation depth meets or exceeds frost depth requirement
- ☐ Footer dimensions (width and depth) match approved plans
- ☐ Rebar size, spacing, and placement meet code requirements
- ☐ Rebar has proper concrete cover (3" min from soil, 2" from forms)
- ☐ All plumbing under slab is installed and positioned correctly
- ☐ Vapor barrier is intact with no tears or unsealed penetrations
- ☐ Forms are clean, braced, and will not move during concrete pour
- ☐ Anchor bolts are on-site and ready to install during pour

- ☐ Site is accessible for inspector vehicle and concrete trucks
- ☐ Building permit is posted in visible location on site
- ☐ Approved foundation plans are available on-site for inspector review

A9: Foundation Inspection

⚠ **STOP — Do not pour concrete until inspection is passed.**

Inspection Requested: _____ Inspector: _____

Inspection Date: _____ Time: _____ Weather: _____

INSPECTION RESULT: ☐ PASSED ☐ FAILED ☐ CONDITIONAL

Inspector Comments / Requirements

- ☐ All inspector-required corrections completed and re-inspected (if needed)

Final Approval Date: _____ Approved to Pour (initials): _____

SECTION B: POUR DAY PREPARATION & EXECUTION**B1: Concrete Order & Scheduling**

	Task	Date	Notes
☐	Calculate concrete quantity needed (cubic yards) — add 5% waste		
☐	Quantity ordered (yd ³): _____ Mix design: _____		
☐	Specify concrete strength (typically 3000–4000 PSI for foundations)		
☐	Request appropriate slump (4–5" typical, 6–7" if using pump)		
☐	Order fiber reinforcement additive (if specified in plans)		

B1: Concrete Order & Scheduling

	Task	Date	Notes
☐	Ready-mix company: _____ Phone: _____		
☐	Pour date scheduled: _____ Arrival time: _____		
☐	Number of trucks: _____ Load spacing (min): _____		
☐	Confirm pump truck if needed: Company: _____ Time: _____		

B2: Weather & Site Conditions

	Task	Date	Notes
☐	Check 3-day weather forecast (no rain during pour, no freeze for 7 days)		
☐	Pour day forecast: High °F: _____ Low °F: _____ Precip? Y / N _____		
☐	Reschedule if temperature will be below 40°F or above 90°F		
☐	Have tarps/plastic sheeting ready in case of unexpected rain		
☐	Have insulated blankets ready if cold weather (below 50°F)		
☐	Verify access road is dry and firm enough for heavy trucks		

B3: Tools, Materials & Labor

Items needed on-site before the pour begins:

Tools & Equipment

- ☐ Wheelbarrows (at least 2) and shovels (multiple)
- ☐ Concrete vibrator (electric or gas-powered)

- ☐ Bull float and hand floats (magnesium or wood)
- ☐ Screed board (straight 2x4 or 2x6, longer than pour width)
- ☐ Hand tamper and concrete rake
- ☐ 4-foot level and tape measure
- ☐ Edging tools and grooving tools (for control joints)
- ☐ Rubber boots and gloves for all workers

Materials

- ☐ Anchor bolts (verify correct size/length, have extras)
- ☐ Curing compound or plastic sheeting for curing
- ☐ Water source for cleaning tools and curing concrete
- ☐ Form release oil (if not already applied)
- ☐ Foam backer rod for control joints (if applicable)

Labor

- ☐ Minimum 4–6 workers lined up and confirmed for pour day
- ☐ Workers briefed on safety (wet concrete is caustic — wear PPE!)
- ☐ Assign specific roles: screeding, floating, vibrating, tool cleaning
- ☐ First aid kit on-site with eye wash (concrete in eyes = emergency!)

B4: During the Pour

	Task	Date	Notes
☐	Pour start time: _____ Truck #1 arrival: _____		
☐	Verify concrete slump before discharge (3–6" typical)		
☐	DO NOT add water to concrete without reducing payment (weakens mix!)		
☐	Pour concrete in sections, working systematically (don't overfill areas)		
☐	Vibrate concrete thoroughly to eliminate voids (especially near forms)		

B4: During the Pour

	Task	Date	Notes
☐	DO NOT over-vibrate (causes segregation and weak concrete)		
☐	Monitor forms for movement, bulging, or failure during pour		
☐	Install anchor bolts while concrete is still workable		
☐	Verify anchor bolt spacing and alignment (check with string line)		
☐	Screed concrete level with top of forms (use straight 2x4)		
☐	Bull float surface to smooth and level		
☐	Create control joints if pouring slab (every 10-15 feet)		
☐	Take concrete test cylinders (if required by inspector/engineer)		
☐	Pour completion time: _____ Total duration (hrs): _____		
☐	Total concrete used (yd ³): _____ Waste/short (yd ³): _____		

SECTION C: POST-POUR FINISHING & CURING**C1: Finishing (if slab/exposed surface)**

	Task	Date	Notes
☐	Wait for bleed water to evaporate (do not finish while water is present!)		
☐	Hand float surface to desired texture		
☐	Edge all perimeter areas with edging tool		
☐	Groove control joints to 1/4 depth of slab		

C1: Finishing (if slab/exposed surface)

	Task	Date	Notes
☐	Broom finish surface if non-slip texture desired		
☐	Do final troweling when concrete is firm enough to support weight		

C2: Curing

	Task	Date	Notes
☐	Apply curing compound OR cover with plastic sheeting (within 12 hours)		
☐	Keep concrete moist for minimum 7 days (14 days in cold weather)		
☐	Protect from freezing temperatures (cover with insulated blankets)		
☐	Protect from excessive heat and sun (mist with water if needed)		
☐	Protect from rain damage in first 24 hours (cover with tarps)		

C3: Form Removal & Waterproofing

	Task	Date	Notes
☐	Wait minimum 3 days before removing forms (7 days in cold weather)		
☐	Remove forms carefully to avoid chipping or damaging concrete		
☐	Remove snap ties and patch holes with mortar/hydraulic cement		
☐	Inspect foundation walls for honeycomb, voids, or defects		
☐	Repair any defects before waterproofing (surface must be sound)		
☐	Apply dampproofing/waterproofing per plan (tar, membrane, etc.)		
☐	Install drainage board or protection board over waterproofing		
☐	Verify perimeter drain tile is in place and functional		

C4: Backfill & Compaction

	Task	Date	Notes
☐	Wait minimum 7 days after pour before backfilling (14 days preferred)		
☐	Ensure waterproofing is fully cured before backfill		
☐	Remove large rocks and debris from backfill material		
☐	Backfill in 12" lifts, compacting each layer (plate compactor)		
☐	Keep heavy equipment away from walls (compact by hand near walls)		
☐	Grade soil away from foundation (2% min slope, 6" drop in 10 feet)		
☐	Protect foundation from erosion until final grading/landscaping		

C5: Final Inspection & Approval

Final Inspection Requested: _____ Inspector: _____

Inspection Date: _____ Time: _____

- ☐ Foundation is fully cured (minimum 7 days since pour)
- ☐ Forms removed and all repairs completed
- ☐ Waterproofing applied and drainage installed
- ☐ Backfill completed and properly compacted
- ☐ Final grading slopes away from foundation

FINAL INSPECTION RESULT: ☐ PASSED ☐ FAILED ☐ CONDITIONAL**Inspector Comments**

Approved to Proceed with Framing (initials): _____ Date: _____

FOUNDATION COMPLETION SIGN-OFF

I certify that the foundation has been constructed according to approved plans and all inspections have passed. The foundation is ready for the next phase of construction.

Owner-Builder Signature: _____ Date: _____

Foundation Contractor (if applicable): _____ Date: _____

PHOTO DOCUMENTATION — FOUNDATION

Document all critical stages with photos. These are essential for warranty claims, inspection appeals, and future reference.

Date	Photo Description	File Name / Number	Notes
	Excavation complete and depth verified		
	Rebar installed before inspection		
	Plumbing under-slab installed		
	Vapor barrier and gravel base		
	Forms ready for inspection		
	During pour — concrete placement		
	Anchor bolts installed		
	Finished surface after pour		
	Forms removed — walls complete		
	Waterproofing applied		
	Drainage system installed		
	Backfill completed		

NOTES, ISSUES & LESSONS LEARNED